

SANA ULLAH AFRIDI

Computer Vision Engineer

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SUMMARY

Computer Vision Engineer with hands-on experience designing and deploying end-to-end detection and segmentation systems. Built a Pakistani license plate recognition system achieving 97.7% mAP50, deployed on Hugging Face with real-time CPU inference. Proficient in YOLOv8, PyTorch, FastAPI, and Docker. 150+ LeetCode problems solved. Actively seeking CV/ML engineering internship roles.

TECHNICAL SKILLS

Computer Vision: YOLOv8, OpenCV, EasyOCR, Object Detection, Semantic Segmentation, Image Classification

Deep Learning: PyTorch, TensorFlow, Keras, Transfer Learning, Data Augmentation, Custom Model Training

ML Libraries: Scikit-learn, NumPy, Pandas, Matplotlib

Deployment: FastAPI, Docker, ONNX, Gradio, Hugging Face Spaces

Tools: Git, GitHub, Roboflow, Jupyter Notebook

Languages: Python, C++

PROJECTS

PK License Plate Scanner [GitHub](#) | [Live Demo](#)

End-to-end Pakistani vehicle license plate detection and OCR system — deployed on Hugging Face Spaces.

- Custom YOLOv8 model trained on Pakistani vehicle images achieving 97.7% mAP50 and 98.7% precision
- Dual-preprocessing pipeline (colour + Otsu thresholding) improves OCR accuracy under variable lighting and contrast
- Regex-based validator for 3 Pakistani plate formats (modern 2-letter, 3-letter, old provincial) filters OCR noise and flags uncertain reads
- CPU-optimised deployment on Hugging Face Spaces — zero GPU cost, live demo accessible publicly

Technologies: Python, YOLOv8, EasyOCR, OpenCV, Gradio, Hugging Face Spaces

Multi-Lane Vehicle Speed Estimation & Violation Detection [GitHub](#)

Real-time computer vision pipeline that tracks vehicles across multiple lanes from a single camera and estimates true speed using perspective-correction geometry, replacing radar-based measurement.

- Built a homography-based perspective-correction pipeline converting pixel displacement into real-world speed (km/h) from a single fixed camera
- Combined YOLOv8 detection with ByteTrack multi-object tracking, point-in-polygon lane assignment, and rolling-window smoothing for stable per-vehicle speed readings
- Automated violation logging (timestamped CSV + cropped snapshots) and a live Streamlit dashboard for per-lane analytics and real-time monitoring
- Diagnosed class-specific detection gaps via confusion-matrix analysis during a custom YOLOv8 fine-tuning experiment on a 6-class local-vehicle dataset

Technologies: Python, YOLOv8, ByteTrack, OpenCV, Streamlit, Homography/Perspective Transform

CutOut — AI Background Remover [GitHub](#) | [Live Demo](#)

Fast background removal web app using a custom-trained YOLOv8n-seg model — deployed as a Dockerised FastAPI app.

- Custom YOLOv8n-seg model trained for 27 epochs on ~2,700 images: 99.4% Box mAP@50, 99.4% Mask mAP@50, 91.6% Mask mAP@50-95

- 4-step pipeline: YOLOv8 segmentation → binary mask → Gaussian edge smoothing → alpha-channel compositing for transparent PNG output
- CPU inference time of 2–4 seconds per image; serves frontend directly via FastAPI as a static HTML file
- Fully Dockerised and deployed on Hugging Face Spaces (free tier) with one-click PNG download

Technologies: Python, YOLOv8n-seg, FastAPI, OpenCV, Pillow, Docker, Hugging Face Spaces

PPE Compliance Detection System [GitHub](#)

Real-time workplace safety monitoring system that detects PPE compliance from video streams.

- Custom YOLOv8 model trained on PPE dataset achieving 85% mAP@50 detecting helmets, vests, and gloves
- Video stream processing pipeline with frame-level non-compliance alerting
- Handles partial occlusion and variable lighting conditions through targeted data augmentation

Technologies: Python, YOLOv8, OpenCV, PyTorch

Weapon Detection System [GitHub](#)

Security-focused firearm detection system for surveillance applications.

- Fine-tuned YOLOv8 on 594-image custom dataset achieving 88.7% precision, 76.3% recall, 86.6% mAP@50
- 4.2ms inference per image on Tesla T4 GPU — suitable for real-time security camera feeds
- Handles variable lighting, angles, and partial occlusion across diverse environments

Technologies: Python, YOLOv8, OpenCV, PyTorch

EDUCATION

BSc Software Engineering — IMSciences, Peshawar

2023 – Present

CERTIFICATIONS

Machine Learning Specialization — Andrew Ng, Coursera

Deep Learning Specialization — Andrew Ng, Coursera

LANGUAGES

English (Fluent) • Urdu (Fluent) • Pashto (Native)